

BenchPress: Benchmarks Preserving Rankings Across Multilingual Post-Training Stages

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BenchPress

Introduction. Post-training requires many recipe decisions (Lambert et al., 2025), validated by evaluation suites that can be long, redundant, and noisy across languages (Singh et al., 2025; Plaza et al., 2024). Heineman et al. (2025) propose a signal-aware framework to determine pretraining benchmark reliability in English. BenchPress extends this work to multilingual post-training stages. We find that the best signal-to-noise definition for (multilingual) mid-training benchmarks is the final checkpoint range over training-step variance, and present the highest SNR benchmarks.

Approach. We aim to find benchmarks with high **Decision Accuracy (DA)**. DA is the fraction of model pairs whose ranking under a cheap proxy still holds under larger compute, either *cross-checkpoint*, *cross-size* and *cross-stage* (pre- vs. post-DPO/RLVR). We search for benchmark characteristics that can serve as a proxy. For each (benchmark, language) cell, we compute **signal**, measuring the spread of scores across models, and **noise**, the variance along training checkpoints of a single recipe. Their ratio is the per-cell SNR. We aim to find the SNR definition that maximizes the Spearman correlation between SNR and each definition of decision accuracy.

Experiments. We run preliminary experiments to validate the methodology for our framework.

Experimental setup. We evaluate 5 intermediate checkpoints for 4 open-weight models across 3 families: smollm3-3b (Bakouch et al., 2025), olmo2-7b/13b, and olmo3-7b (Team Olmo et al., 2025), evaluated with lm-evaluation-harness (Gao et al., 2024) on RCP. 30 benchmarks used include: ARC (Clark et al., 2018), HellaSwag (Zellers et al., 2019), HumanEval-Instruct (Chen et al., 2021), MGSM-CoT (Shi et al., 2023; Wei et al., 2022), GSM8K-CoT (Cobbe et al., 2021), IFEval (Dussolle et al., 2025), TruthfulQA-MC2 (Lin et al., 2022), across {en, hi, es, ar, ru} when

available. We calculate SNR under multiple definitions, as mentioned in Heineman et al. (2025).

Initial results. The two SNR definitions most correlated with DA reach $\rho = +0.84$. In both, signal is the spread of final-checkpoint scores (measured by range or dispersion) and the noise is the training-step variance; all variants are ranked in Figure 1. Under this definition, ARC-e and its multilingual variants, all HellaSwag tasks, and HumanEval-Instruct are the highest-SNR benchmarks (Figure 2). DA vs per-cell SNR is shown in Figure 3, and checkpoint scores across multiple benchmarks are in Figure 4.

Challenges and mitigations. First, queue latency and the combinatorial explosion of model $\times$ benchmark runs, addressed by focusing on a smaller set for the preliminary results. Second, we evaluate only mid-training checkpoints since post-training intermediate checkpoints are rarely released publicly. To address this limitation, we plan to set up an SFT run to post-train models using a subset of Tulu-3 (Lambert et al., 2024) using both full fine-tuning and LoRA (Hu et al., 2022).

Next Steps. To finalize the project, we plan to: (1) Broaden the SNR design space with new signal and noise definitions (e.g., training seed, prompt-format, temperature noise). (2) Post-train midtraining checkpoints of open-weight models and extend the SNR analysis to the post-training stage. (3) Report, per DA form, language, and stage, the SNR variant that best predicts DA under Spearman correlation. (4) Propose a benchmark subset selection method based on sample-level SNR. (5) Study the framework generalization to benchmarks that evaluate different capabilities and to typologically diverse languages {zh, ja, sw, bn, te}. *Risks:* Post-training will add compute load to our experiments, so we start with one family and a short recipe. We will first define the methodological framework and launch runs progressively.

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A Appendix

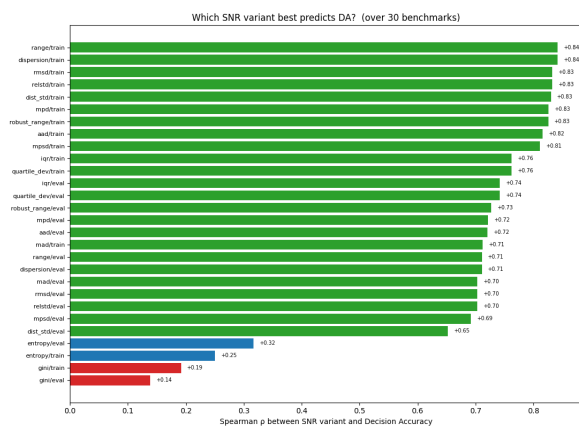


Figure 1: Spearman correlation between each candidate SNR definition and pairwise Decision Accuracy. The *range/train* and *dispersion/train* pairs both reach $\rho \approx 0.84$. We use *range/train* for the rest of the analysis.

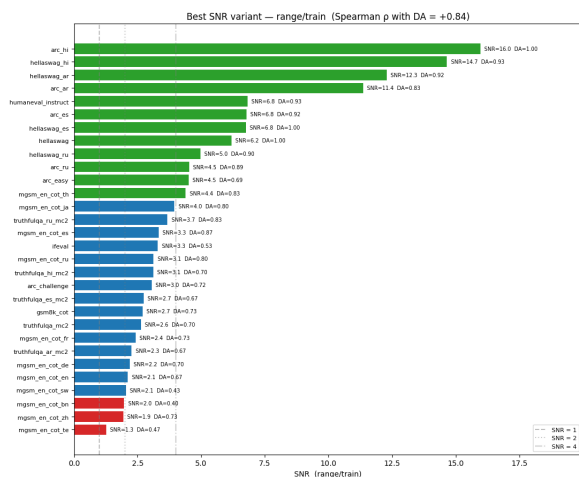


Figure 2: Per-benchmark SNR across the evaluated languages. ARC-easy, HellaSwag and HumanEval-Instruct form the high-SNR diagnostic tier; low-SNR cells flag noisy or saturated evaluations.

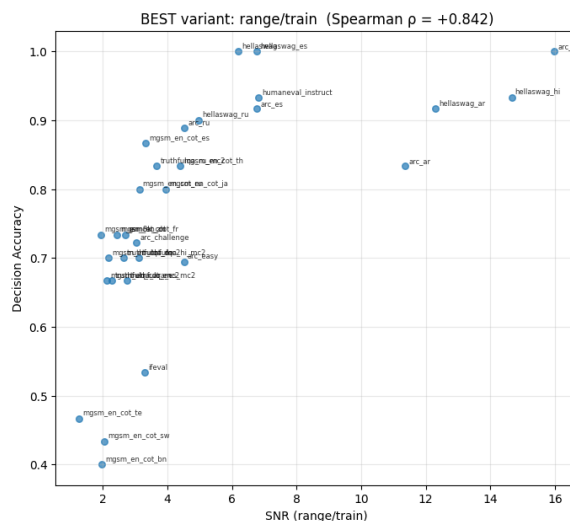


Figure 3: Per-cell SNR against Decision Accuracy across (benchmark, language) pairs. The positive trend validates SNR as a cheap predictor of ranking-preserving benchmarks.

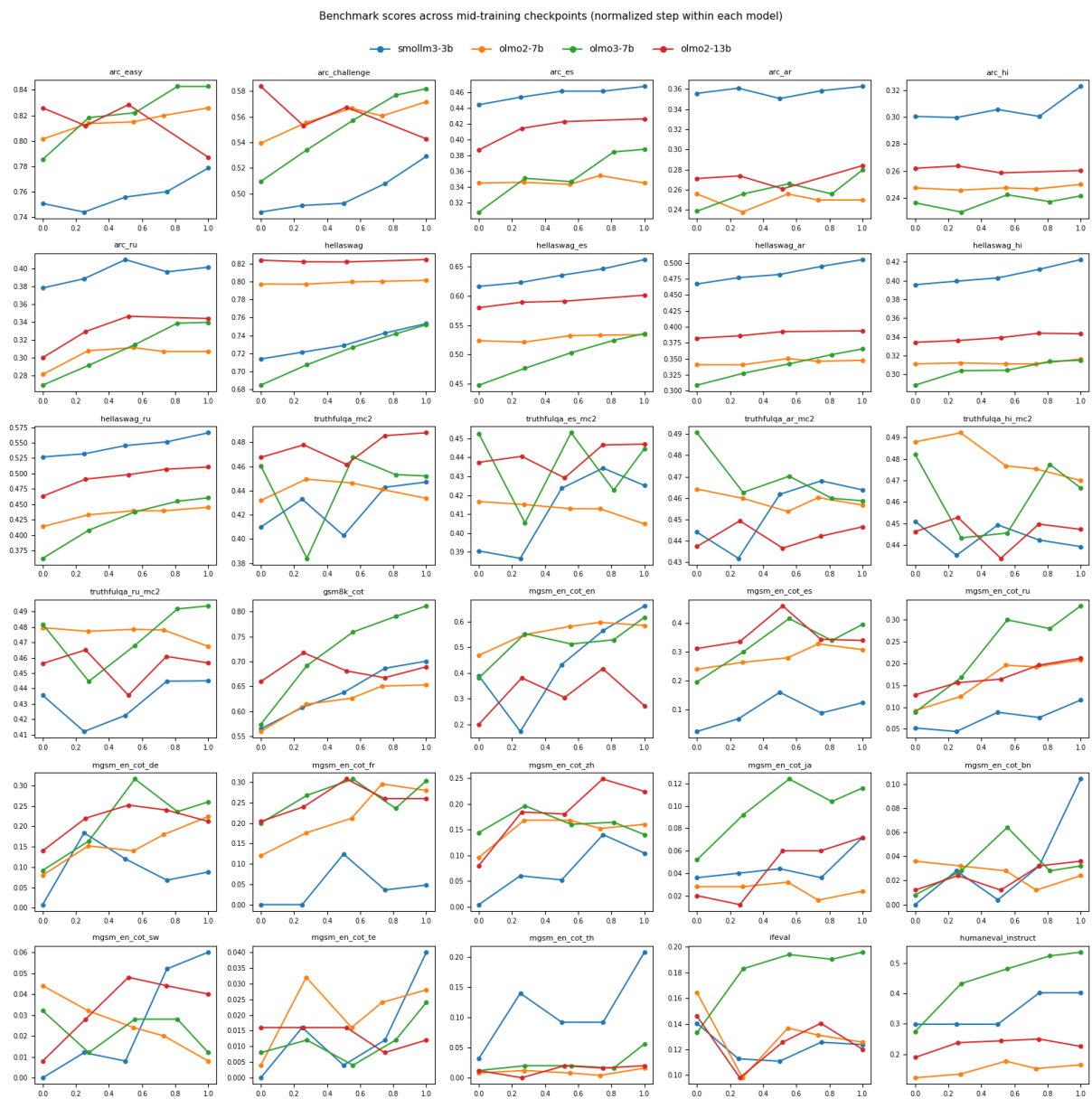


Figure 4: Benchmark scores across all models and checkpoints.